

VibroBP: An Oscillometric Smartphone Blood Pressure App

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ABSTRACT

This interactivity features a smartphone-based method for measuring blood pressure (BP) using the oscillometric method. To measure BP, it is necessary to measure (1) the pressure applied to the artery and (2) the local blood volume change. This is accomplished by performing an oscillometric measurement at the finger's digital artery, whereby a user presses down on the phone's camera with steadily increasing force. The camera is used to capture the blood volume change using photoplethysmography. We devised a novel method for measuring the force applied by the finger without the use of specialized smartphone hardware with a technique called Vibrometric Force Estimation (VFE). The fundamental concept of VFE relies on a phenomenon where a vibrating object is dampened when an external force is applied on it. This phenomenon can be recreated using the phone's own vibration motor, where the resulting damped vibration is measured using the smartphone's Inertial Measurement Unit (IMU). A cross device reliability study with three smartphones of different manufacturers, shape, and prices results in similar force estimation performance across all smartphone models. In an N=24 validation study of the BP measurement, the smartphone technique achieves a MAE of 8.4 mmHg and 8.5 mmHg of systolic and diastolic BP, respectively, compared to an FDA approved BP cuff. The mean error (bias) and standard deviation are 0.44 ± 10.2 mmHg and 0.03 ± 10.5 mmHg for systolic and diastolic, respectively. The vision for this technology is not necessarily to replace existing BP monitoring solutions, but rather to introduce a downloadable smartphone software application that could serve as a low-barrier hypertension screening measurement fit for widespread adoption.

CCS CONCEPTS

• Human-centered computing → Smartphones; • Applied computing → Consumer health.

KEYWORDS

Smartphone, Blood Pressure, Health

ACM Reference Format:

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1 INTRODUCTION

The VibroBP application enables a novel oscillometric smartphone BP measurement method that requires no hardware modifications, no physical attachments, nor any need for individual calibration with cuff-based BP measurements. The proposed measurement can be enabled solely with software by downloading a smartphone application on nearly any smartphone. It is NOT an “optical” BP measurement relying on pulse transit time (PTT), pulse arrival time (PAT), pulse wave analysis (PWA), or any other technique estimating the relative change in BP. The application performs absolute BP measurements based on the oscillometric method similar to that employed by automated BP cuffs, in accordance with the recommendation of the American Medical Association (AMA) and the American Heart Association (AHA) [11].

The oscillometric method requires measuring two key metrics: applied force and local blood volume of an artery. Local blood volume of an artery can be accurately estimated via photoplethysmography (PPG) with the smartphone camera. Measuring applied force with no attachments serves as the key insight presented in the paper to enable a novel BP measurement. Figure 1 depicts these key concepts of smartphone oscillometric BP measurements proposed in this paper. To measure applied force, we propose a force estimation technique utilizing only the vibration motor and inertial measurement unit (IMU). Nearly every phone contains an IMU with an accelerometer and gyroscope to measure the linear acceleration and angular velocity that enables phone position inferences used for rotating the screen, playing games, counting steps, gesture recognition, and much more. Similarly, nearly all smartphones contain a vibration motor to provide haptic feedback for messages, phone calls, and screen interactions. Taking advantage of the pre-existing vibration motor and IMU allows us to create a model to measure force. The phone is set to maximally vibrate while the IMU measures the motion as the phone oscillates (from the vibration). When force is applied during vibration, the IMU signal changes as the oscillations are altered or dampened. The alterations in the vibration signal allows us to train a model to estimate the applied force. We refer to this method of estimating applied force using the

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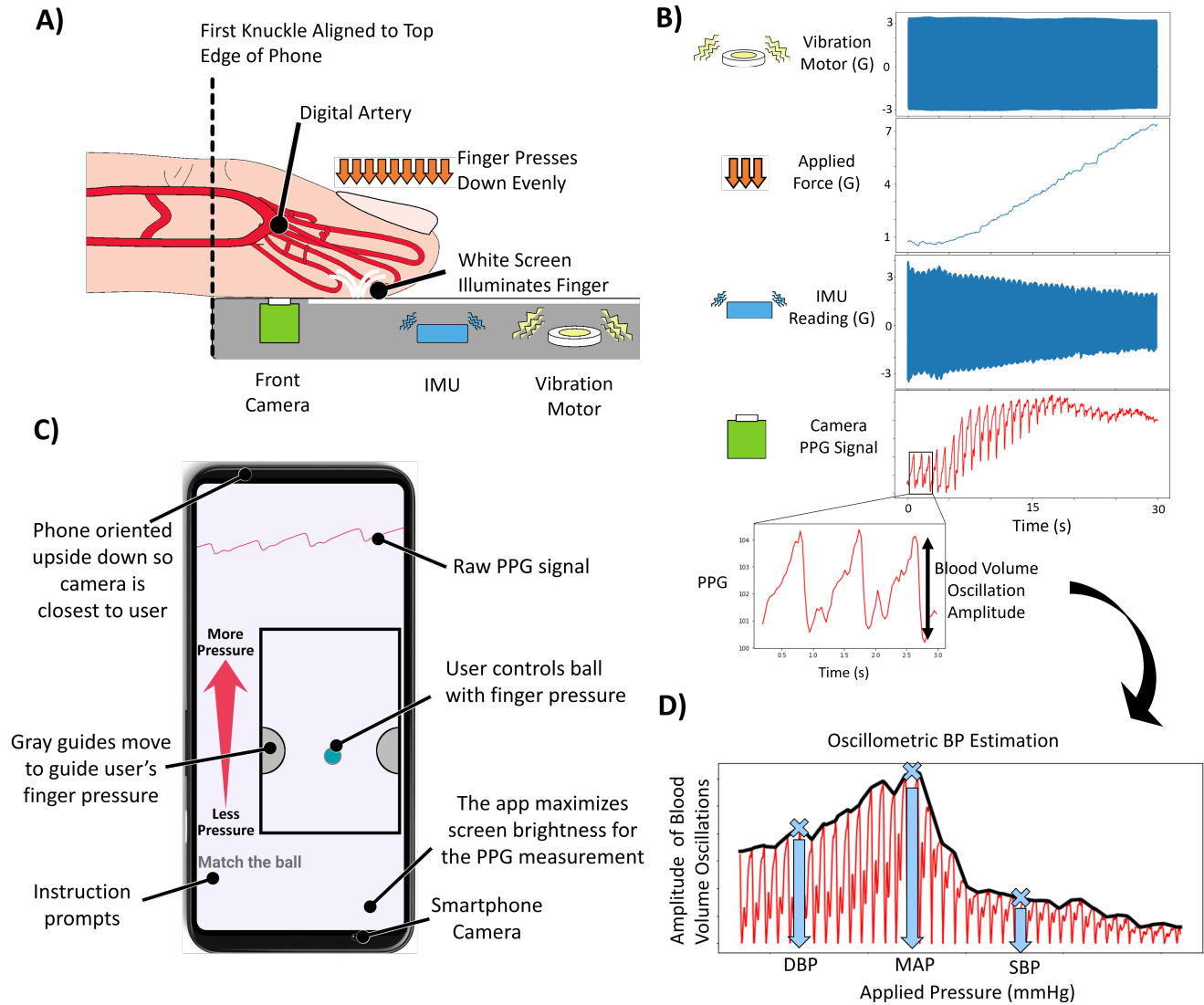


Figure 1: (A) Simplified diagram of the smartphone vibrometric force estimation, where the vibration motor drives oscillatory motion and the finger applies a force. The smartphone Inertial Measurement Unit (IMU) records the motion and the smartphone camera records PPG. (B) Applied finger force and linear y-axis acceleration of smartphone IMU are plotted vertically. As force increases, the oscillation amplitude of the y-axis acceleration decreases. The raw PPG signal is also plotted vertically to show how increased force affects the blood volume. (C) Smartphone app with visual queue for finger, guideline for applied force, and plot of PPG signal. (D) Amplitude of blood volume oscillations plotted against applied finger pressure. The approximate points of Diastolic Blood Pressure (DBP), Mean Arterial Pressure (MAP), and Systolic Blood Pressure (SBP) are indicated for interpretability.

vibration motor and IMU signal as Vibrometric Force Estimation (VFE).

This work represents the first absolute BP measurement technique on a smartphone without special components or attachments. Most prior research into smartphone and smartwatch BP measurements perform relative BP measurements using PTT or PWA, but these techniques are fundamentally different. PTT and PWA only provide relative change in BP. These measures of relative change

assume an individual has access to a BP cuff to calibrate their individual measurement due to the model's reliance on arterial stiffness and arm length, otherwise, the relative measurements cannot be meaningfully interpreted [5–10, 12, 13]. Prior works on absolute smartphone BP measurement require custom built attachments to measure force [1, 16]. One prior work functions only on a specific smartphone model with a built in force sensitive screen, which is

not a common feature [2]. Our proposed method is the first smartphone BP measurement that requires no attachment and functions on nearly any smartphone model, only requiring a vibration motor, camera, and IMU.

2 METHODS

2.1 Vibrometric Force Estimation

For each smartphone, data from the force dampening effect profiling study provides IMU data during a range of applied force values. To estimate force, the IMU profile is used to train a machine learning model for each phone model. The IMU data contains three axis accelerometer data and three axis gyroscope data. Each axis of the IMU data contains high frequency signals from the vibration motor as well as noise at lower frequencies. The applied force from the finger most significantly affects the amplitude of the signal oscillation. To leverage this understanding, the rolling standard deviation of each axis of the accelerometer and gyroscope data serves as a feature. The features most significantly correlated to the force data are included as model inputs, while other features are excluded. With these IMU data features as inputs, a regression model (multivariate linear regression model or gradient boosting) is trained using a ten percent holdout for validation. This process is repeated for each smartphone such that a distinct force estimation model exists for each smartphone model.

2.2 Smartphone Camera PPG

For the oscillometric BP measurement, we leverage the smartphone camera PPG to sense volumetric changes in blood flow at the transverse palmar arch branch of the digital artery. As the volume of blood flowing through the artery near the fingernail bed changes, the reflective properties of the tissue in the region changes. As such, the smartphone camera can record changes in blood volume by measuring the changes in the reflective properties.

In the study, the phone screen is set to a pure white background with maximal brightness and the user is instructed to place their index finger over the camera (as aided by a visual signifier). The bright white screen illuminates the finger and the camera records pixel intensity changes in the red channel as a proxy for blood volume changes. To account for skin tone and lighting variations, the smartphone performs the calibration outlined in Xuan and Barry et. al. [16] with the user's finger over the camera. This calibration protects against under- or over-saturated PPG measurements.

2.3 Oscillometric Blood Pressure Estimation

During the smartphone BP measurement, the smartphone camera records the PPG signal while the user exerts force with the index finger on the camera. Applying force causes changes in the PPG signal to create an oscillogram shape. When the applied force is greater than the diastolic BP, the blood volume during the systolic pulse phase increases and the blood volume in the diastolic phase decreases. The total volume of blood flowing through the artery is approximately the same, but pressure on the artery causes more blood to flow through at the systolic pulse phase and less at the diastolic pulse phase. This increases the prominence of the PPG signal. When the applied force surpasses the mean arterial pressure, the blood volume begins to be limited even during the systolic

phase of the pulse. As such, the systolic phase PPG prominence begins to decrease and the blood volume during the diastolic phase approaches zero. The PPG prominence continues to decrease even after the applied force surpasses the systolic BP. As the pressure increases, the signal will decrease to zero (or noise) as the blood flow through the artery during both the systolic and diastolic phase approaches zero.

A plot of the pulse prominence through the range of applied forces (from less than diastolic BP to greater than systolic BP) creates a Gaussian or skew Gaussian-like shape that we refer to as the oscillogram. The oscillometric method utilized by most FDA approved BP cuffs and numerous prior academic works utilize the oscillogram shape to estimate BP. Our method of estimating BP relies on the features and findings identified in prior work [3].

We generate features by focusing on features and concepts previously identified in prior work [3] for oscillometric BP measurements. The following features are included in the BP estimation model: the parameters for the piecewise Gaussian of best fit, the applied force at the maximum prominence value, average prominence at low force values, maximum normalized prominence, and the applied force at the maximum of the Gaussian of best fit.

These key features of the oscillometric method are the only inputs into our BP prediction model. Note that the attributes related to exercise including heart rate and pulse shape are not included in the features provided to the model. Additionally, indirect measurements of heart rate or pulse are further avoided by interpolating all oscillograms to have the same number of points with the same force values. With the key features, we trained a least absolute shrinkage and selection operator (LASSO) regression model to predict systolic and diastolic measures of BP [14]. The model is trained and tested using a hold one measurement out validation. In this training scheme, the model is trained on all data excluding the one or more BP measurements. The model is then used to predict the BP of the holdout measurement. This process is repeated for each measurement.

3 RESULTS

3.1 Blood Pressure Proof-of-Concept Study

The BP proof-of-concept study comparing the smartphone to the FDA-approved BP cuff with a hold one measurement out validation achieved a mean absolute error of 8.4 mmHg and 8.5 mmHg for systolic and diastolic, respectively. The mean error (bias) and standard deviation are 0.44 ± 10.2 mmHg and 0.03 ± 10.5 mmHg for systolic and diastolic, respectively. The Pearson correlation coefficient was 0.77 and 0.58 for systolic and diastolic measurements, respectively. The BP cuff measurements ranged from 87 to 165 mmHg systolic and 54 to 119 mmHg for diastolic. These results are depicted in Figure 2.

In evaluating the proof of concept more specifically around hypertension screening, we categorize BP values based on the AMA/ACC standards [15], which involves determining if an individual is below 120mmHg (normal), between 120-130 mmHg (elevated), and above 130 mmHg (hypertensive). Note that the confusion matrix results use a scalar shift. In categorizing the model results as normal (below 130mmHg systolic BP) or hypertensive

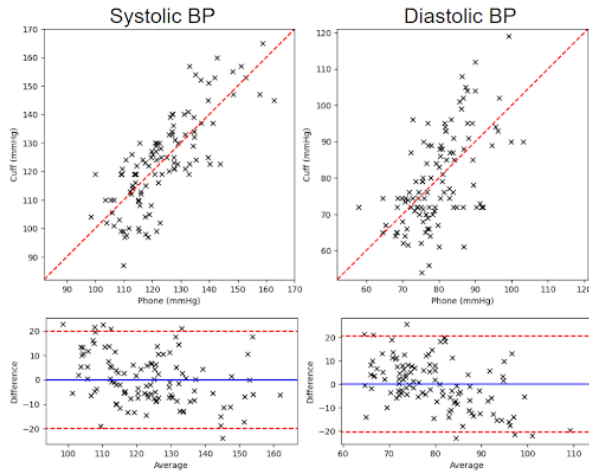


Figure 2: Feasibility Study Results

(above 130mmHg systolic BP), the ROC analysis reveals an area under the curve (AUC) score of 0.90 for detecting high BP individuals, characterized by systolic BP greater than 130 mmHg.

3.2 Blood Pressure Study Participants

The full BP validation study recruitment includes $N=34$ participants, each with at least three resting smartphone BP measurements. During the study, $N=33$ participants correctly applied force in at least 1 of their measurements as guided by the smartphone application. $N=22$ participants contained at least 1 usable measurement (valid force and ppg signal combination) for BP prediction. The participant population is 27% White, 62% Asian, and 11% Hispanic, consisting of 50% males and 50% females. The mean age of the participants is 25.6 years with a standard deviation of 4.3. The average resting systolic and diastolic BP is 103mmHg and 70mmHg, respectively.

3.3 Skin Tone and Ethnicity

An ANOVA test is conducted to determine the impact of skin tone and ethnicity on performance. For both skin tone and ethnicity, the group variations are not statistically significant with p-values of 0.32 and 0.25, respectively.

3.4 Cross Device Validity

We ran a benchtop experiment demonstrating the force estimation technique across multiple models of phones. The accuracy of measuring applied finger force using VFE is evaluated across three smartphones: Google Pixel 4, Samsung Galaxy A53, and Motorola Moto G Power. The mean absolute error, standard deviation, bias, and correlation coefficient for the Google Pixel 4 is 0.51N, 0.34N, -0.49N, and 0.98, respectively. The mean absolute error, standard deviation, bias, and correlation coefficient for the Samsung Galaxy A53 is 0.55N, 0.65N, -0.18N, and 0.95 respectively. The mean absolute error, standard deviation, bias, and correlation coefficient for the Motorola Moto G Power 2022 is 0.39N, 0.53N, -0.02N, and 0.96, respectively. The average correlation coefficient across all phones is 0.97.

3.5 Discussion

The study results demonstrate the feasibility of oscillometric smartphone BP measurements using VFE and PPG measurements. The systolic correlation coefficient of 0.78 and an MAE of 8.4 mmHg are comparable to state-of-the-art smartphone-based BP measurements performed with attachments or custom equipment [1, 16]. Additionally, the results suggest the smartphone measurement functions equally for groups of different ethnicities or skin tones. The central improvement offered in the work is the method for performing absolute BP measurements that can be applied across smartphone models without modifications or attachments.

Despite the promise, it is important to recognize the limitations of the proposed technology. The performance of the diastolic BP prediction is limited with a correlation coefficient of 0.42. This suggests that isolated diastolic hypertension could not be well detected; however, systolic BP alone can still accurately screen for hypertension in all but 3.2 % of the population because the prevalence of isolated diastolic hypertension is extremely low compared to other types of hypertension [4]. Future exploration of data collection at lower force levels and features more specific to diastolic BP will likely improve the diastolic BP performance.

The VFE technique is smartphone dependent. However, the calibration can be performed relatively quickly and once the profiling is completed for a phone model, VFE can work on all phones of that model. It would be expected that the application developer would perform the calibrations so the end user can simply download a ready-to-use application.

4 CONCLUSION

While the overall goal of this interactivity is to present the novel smartphone BP measurement methodology using vibrometric force estimation, there is significant potential for future work. The proof-of-concept studies performed here demonstrate feasibility, but future studies involving larger scale data collections of participants with varied hypertensive conditions will be performed to further develop vibrometric force estimation model parameters, train larger models, and better evaluate accuracy. Additionally, the studies performed thus far have all been performed under supervision in a controlled lab environment. There is a need for future work to explore the usability of the system for different demographics, especially focusing on the potential for individuals to perform measurements in at-home environments. Future research should include investigations of how conditions like diabetic neuropathy, Parkinson's disease, or thick skin calluses might affect the measurement.

REFERENCES

- [1] Anand Chandrasekhar, Chang-Sei Kim, Mohammed Naji, Keerthana Natarajan, Jin-Oh Hahn, and Ramakrishna Mukkamala. 2018. Smartphone-based blood pressure monitoring via the oscillometric finger-pressing method. *Science Translational Medicine* 10, 431 (March 2018), eaap8674. <https://doi.org/10.1126/scitranslmed.aap8674>
- [2] Anand Chandrasekhar, Keerthana Natarajan, Mohammad Yavarimanes, and Ramakrishna Mukkamala. 2018. An iPhone Application for Blood Pressure Monitoring via the Oscillometric Finger Pressing Method. *Scientific Reports* 8, 1 (Dec. 2018), 13136. <https://doi.org/10.1038/s41598-018-31632-x>
- [3] Anand Chandrasekhar, Mohammad Yavarimanes, Jin-Oh Hahn, Shih-Hsien Sung, Chen-Huan Chen, Hao-Min Cheng, and Ramakrishna Mukkamala. 2019. Formulas to Explain Popular Oscillometric Blood Pressure Estimation Algorithms. *Frontiers in Physiology* 10 (Nov. 2019), 1415. <https://doi.org/10.3389/fphys.2019.01415>

- [4] Clara K. Chow. 2013. Prevalence, Awareness, Treatment, and Control of Hypertension in Rural and Urban Communities in High-, Middle-, and Low-Income Countries. *JAMA* 310, 9 (Sept. 2013), 959. <https://doi.org/10.1001/jama.2013.184182>
- [5] Alair Dias Junior, Srinivasan Murali, Francisco Rincon, and David Atienza. 2016. Methods for reliable estimation of pulse transit time and blood pressure variations using smartphone sensors. *Microprocessors and Microsystems* 46 (Oct. 2016), 84–95. <https://doi.org/10.1016/j.micpro.2016.06.001>
- [6] Lorenz Frey, Carlo Menon, and Mohamed Elgendi. 2022. Blood pressure measurement using only a smartphone. *npj Digital Medicine* 5, 1 (July 2022), 86. <https://doi.org/10.1038/s41746-022-00629-2>
- [7] Alexandre Joosten, Alexandra Jacobs, Olivier Desebbe, Jean-Louis Vincent, Saxena Sarah, Joseph Rinehart, Luc Van Obbergh, Alexander Hapfelmeier, and Bernd Saugel. 2019. Monitoring of pulse pressure variation using a new smartphone application (Capstesia) versus stroke volume variation using an uncalibrated pulse wave analysis monitor: a clinical decision making study during major abdominal surgery. *Journal of Clinical Monitoring and Computing* 33, 5 (Oct. 2019), 787–793. <https://doi.org/10.1007/s10877-018-00241-4>
- [8] Alair Dias Junior, Srinivasan Murali, Francisco Rincon, and David Atienza. 2015. Estimation of Blood Pressure and Pulse Transit Time Using Your Smartphone. In *2015 Euromicro Conference on Digital System Design*. IEEE, Madeira, Portugal, 173–180. <https://doi.org/10.1109/DSD.2015.90>
- [9] He Liu, Kamen Ivanov, Yadong Wang, and Lei Wang. 2015. Toward a Smartphone Application for Estimation of Pulse Transit Time. *Sensors* 15, 10 (Oct. 2015), 27303–27321. <https://doi.org/10.3390/s151027303>
- [10] Ramakrishna Mukkamala, Jin-Oh Hahn, Omer T. Inan, Lalit K. Mestha, Chang-Sei Kim, Hakan Toreyin, and Survi Kyal. 2015. Toward Ubiquitous Blood Pressure Monitoring via Pulse Transit Time: Theory and Practice. *IEEE Transactions on Biomedical Engineering* 62, 8 (Aug. 2015), 1879–1901. <https://doi.org/10.1109/TBME.2015.2441951>
- [11] Paul Muntner, Daichi Shimbo, Robert M. Carey, Jeanne B. Charleston, Trudy Gaillard, Sanjay Misra, Martin G. Myers, Gbenga Ogedegbe, Joseph E. Schwartz, Raymond R. Townsend, Elaine M. Urbina, Anthony J. Viera, William B. White, Jackson T. Wright, and on behalf of the American Heart Association Council on Hypertension; Council on Cardiovascular Disease in the Young; Council on Cardiovascular and Stroke Nursing; Council on Cardiovascular Radiology and Intervention; Council on Clinical Cardiology; and Council on Quality of Care and Outcomes Research. 2019. Measurement of Blood Pressure in Humans: A Scientific Statement From the American Heart Association. *Hypertension* 73, 5 (May 2019). <https://doi.org/10.1161/HYP.0000000000000087>
- [12] C.C.Y. Poon and Y.T. Zhang. 2005. Cuff-less and Noninvasive Measurements of Arterial Blood Pressure by Pulse Transit Time. In *2005 IEEE Engineering in Medicine and Biology 27th Annual Conference*. IEEE, Shanghai, China, 5877–5880. <https://doi.org/10.1109/IEMBS.2005.1615827>
- [13] Patrick Schoettker, Jean Degott, Gregory Hofmann, Martin Proença, Guillaume Bonnier, Alia Lemkaddem, Mathieu Lemay, Raoul Schorer, Urvan Christen, Jean-François Knebel, Arlene Wuerzner, Michel Burnier, and Gregoire Wuerzner. 2020. Blood pressure measurements with the OptiBP smartphone app validated against reference auscultatory measurements. *Scientific Reports* 10, 1 (Oct. 2020), 17827. <https://doi.org/10.1038/s41598-020-74955-4>
- [14] Robert Tibshirani. 2011. Regression shrinkage and selection via the lasso: a retrospective: Regression Shrinkage and Selection via the Lasso. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 73, 3 (June 2011), 273–282. <https://doi.org/10.1111/j.1467-9868.2011.00771.x>
- [15] Paul K. Whelton, Robert M. Carey, Wilbert S. Aronow, Donald E. Casey, Karen J. Collins, Cheryl Dennison Himmelfarb, Sondra M. DePalma, Samuel Gidding, Kenneth A. Jamerson, Daniel W. Jones, Eric J. MacLaughlin, Paul Muntner, Bruce Ovbiagele, Sidney C. Smith, Crystal C. Spencer, Randall S. Stafford, Sandra J. Taler, Randall J. Thomas, Kim A. Williams, Jeff D. Williamson, and Jackson T. Wright. 2017. ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension* 71, 6 (2017). <https://doi.org/10.1161/HYP.0000000000000065>
- [16] Yanan Xuan, Colin Barry, Jessica De Souza, Jessica H. Wen, Nick Antipa, Alison A. Moore, and Edward J. Wang. 2023. Ultra-low-cost mechanical smartphone attachment for no-calibration blood pressure measurement. *Scientific Reports* 13, 1 (May 2023), 8105. <https://doi.org/10.1038/s41598-023-34431-1>